Gauge crossover: comparison tools and their missing estimates

Maintained conclusion. The examined comparison and finite-block techniques provide possible ingredients for crossover control, not an exhaustive list of ways to prove a physical continuum gap. A normalized, volume-uniform mixing estimate must be established for the actual generated interaction. Neither a loop monotonicity statement nor a finite certification proposal supplies it by itself. Weak-to-strong transport, cutoff removal and physical reconstruction remain additional steps.

Correlation inequalities: what would actually transfer

If Wilson loop expectations increase with β_W, an area-law string tension defined by the corresponding limit is nonincreasing where the limit and comparison are valid. A lower area-law bound can then transfer from a weaker to a stronger coupling. Monotonicity of a loop expectation alone does not imply monotonicity of a connected correlator, its decay exponent or the vacuum-sector spectral gap. Both terms of a truncated correlator may increase. The old claim that this closes every stronger-coupling channel is withdrawn.

Ginibre's method needs the interaction to lie in a cone of products of single-site functions satisfying the duplicate-variables condition. For the character cone of $SU(2)$ the single-site condition reduces to

$$\iint \prod_{i=1}^{n} \left(\chi_i(x) - \chi_i(y)\right) dx\, dy = \sum_{S \subseteq \{1..n\}} (-1)^{|S^c|}\, N_S\, N_{S^c}, \qquad N_S = \dim \mathrm{Hom}\left(1, \bigotimes_{i \in S} V_i\right),$$

which vanishes for odd n, equals $2N_{1234} + 2(\delta_{12}\delta_{34} + \delta_{13}\delta_{24} + \delta_{14}\delta_{23}) \geq 0$ for $n = 4$, and is positive in every case computed at $n = 6$ (six spin $\frac{1}{2}$: 70; six spin 1: 100; four spin $\frac{1}{2}$ and two spin 1: 28; six spin 2: 260). The single-site condition is therefore no obstruction as far as checked. The obstruction is that the plaquette term $\mathrm{Re}\,\mathrm{tr}(U_1U_2U_3U_4)$ is not a nonnegative combination of products of single-link functions: the abelian proof writes $\cos(\theta_1 + \dots)$ through sums and differences of angles, which uses the commutativity of the group, and no analogue exists for a matrix product. The second Griffiths inequality was not established by the sources read here for $SU(2)$ and $SU(3)$ lattice gauge theory, ; no proof or counterexample is supplied here, and the centre-based inequalities that do exist are listed in the what-would-unblock note §1.

The finite checks are not an all-order character-cone theorem. The obstacle identified here is the absence of the needed positive plaquette decomposition in the examined construction, not a proof that every correlation inequality for a nonabelian theory is impossible.

Tools and source scope

These are the sources previously cited in this family, at **metadata level** unless a linked canonical proof records a deeper read. They are leads and scope comparisons, not independently verified universal impossibility claims.

Tool/source	Potential input	Missing transfer
Ginibre, CMP 16 (1970) 310; Aizenman–Barsky–Fernández, JSP 47 (1987) 343	Positive cones and Ising-type sharpness	A nonabelian plaquette interaction satisfying the required hypotheses
Mack–Petkova, Ann. Phys. 123 (1979) 442 and 125 (1980) 117; Tomboulis–Yaffe, CMP 100 (1985) 313	Centre, electric-flux and vortex/loop comparisons	A vacuum-sector physical spectral bound; a string tension is a different observable

Tool/source	Potential input	Missing transfer
Fröhlich–Israel–Lieb–Simon, CMP 62 (1978) 1	Reflection-positive chessboard bounds for specified events	Good-region mixing and a normalized block free-energy estimate
Fröhlich–Simon–Spencer, CMP 50 (1976) 79	Gaussian-domination/infrared techniques for their stated spin interactions	The needed domination for the nonquadratic plaquette product; nonabelian target curvature alone is not the failure reason
Knabe, JSP 52 (1988) 627	A finite-size gap criterion under frustration-free hypotheses	Those hypotheses are not verified for interacting KS at general coupling; this does not exhaust all finite-size criteria
Finite mixing proposal, Dobrushin proof	A certified supremum over boundary conditions could imply a Wilson-transfer gap	No certificate at the physical crossover or for the generated interaction is supplied
Exact flow conjugation	Spectrum-preserving change of variables	New effective decomposition, relative error and iterative control
Small-volume bridge	Compressed zero-mode estimates with a named complement/Schur target	Renormalized cutoff-uniform transfer and passage to large physical volume

A signed spin-foam representation does not make the original Wilson probability weight negative, and positive plaquette coefficients do not make every recoupling amplitude positive. Centre comparisons, electric-flux sectors and glueball correlations must name the sector they control. The 2007 approximate centre-decimation proposal (arXiv:0707.2179, metadata) was not established as a theorem by this repository's reading.

Gauge fixing can change a conditional specification; merely subtracting a maximal-tree fraction from a neighbour count does not prove a new Dobrushin coefficient. The earlier numerical suggestion $444 \mapsto 330$ is unsupported here. Likewise, neither a perturbative four-dimensional Langevin construction nor its obstruction is proved by this source survey; an auxiliary stochastic clock is not a physical energy gap.

Finite certification and the actual interaction

The earlier illustrative boxes at $\beta_W \simeq 6$ had roughly 216–2000 links, hence thousands of integration variables. These are scale estimates from the bands note, not a complexity theorem excluding every certification method. Monte Carlo by itself is not a certified supremum over boundary conditions. A rigorous polymer enumeration needs an analytic tail and its own convergence region. The explicit Wilson criteria here prove sufficient strong-coupling regions, not limits on all possible future methods. The strong-coupling proofs and Dobrushin criterion own their constants.

The weak side has explicit but restricted one-step tests in Gaussian fluctuation control and decay/remainder thresholds. The mid-plane notebook later proves different retained-barrier weak-strip estimates. Their coefficients and cutoff/volume hypotheses cannot be silently substituted for a physical-coupling full integral.

After blocking, all generated interactions must be controlled. There is no proved arrival in the completely analytic set at order-one coupling and no proved universal statement that all difficulty is one fixed unknown lattice interaction. A possible mixing certificate at that

scale would still need stable transport from the weak trajectory, physical spectral visibility, nontriviality and continuum reconstruction.

Consequence for STATE

The productive questions are the exact normalized estimates still missing, listed in the gauge atlas. The older claim of exactly two unblocking routes, the unsupported closed large-field tail and flow sharpness, and the assertion that all other methods fail are withdrawn. The retained source evidence explains what each examined method would require; it changes no open target into a theorem and initiates no new research during consolidation.